

CS425 Fall 2026 – Homework 2

(a.k.a. “Once upon a time in Distributed Hollywood”)

Out: Sep 21, 2026. Due: Oct 4, 2026 (Sunday, 11:59 pm US Central)

Topics: P2P Systems, Key-value Stores, Time and Ordering

Instructions:

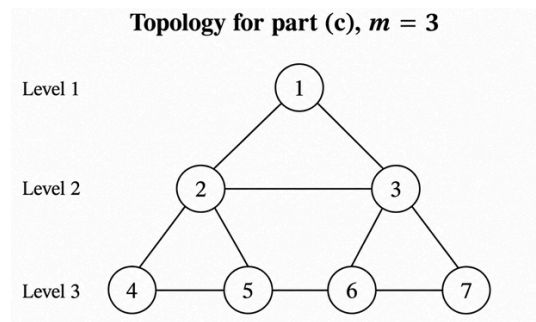
1. Please double-check that the top of this page mentions the correct semester you are taking the course in (otherwise, you are looking at an old version and we will not accept your submitted solutions).
2. Attempt any **8 out of the 10 problems** in this homework. If you attempt more, we will grade only the first 8 solutions that appear in your homework. Each problem has the same grade value (10 points each).
3. Your **solutions must be typed and not handwritten**. Figures (e.g., timelines) and equations (if any) may be hand-drawn (and scanned).
4. Submit PDF only. **Start each problem on a fresh page.**
5. Please submit on Gradescope. [<https://www.gradescope.com/>] Self-signup using the code on the website.
6. **Important: On Gradescope, please tag each page with the problem number!**
7. Homework is **due on the date and time noted above. No extensions.**
For DRES students only: once the solutions are posted (typically a few hours after the HW is due), subsequent submissions will get a zero.
8. Always justify all your answers, and show your work.
9. Unless otherwise specified in the question, the only resources you can avail of in your HW are the provided course materials (slides, textbooks, etc.) and communication with instructors/TA via the discussion forum.
10. You can discuss lecture concepts and the questions on Piazza and with your friends, but do not discuss solutions or ideas on Piazza.

Prologue: You have just been made the technical head in a production company that is producing a new Hollywood movie. The movie is sure to be a blockbuster, with a lot of well-known actors and actresses hired to star in it. Amazingly, many of them know distributed systems! You run into them every day on the set. Here is what ensues. All characters and their actions in this homework are meant to make the homework fun! Any resemblance of their actions or opinions to real events or places is purely coincidental. All these stories are fictional.

Problems:

1. Walt Disney and Pokemon want to build a virtual theme park where young customers will all wear VR goggles and be connected via a Gnutella P2P system. At one point in time, the Gnutella topology is a perfectly balanced binary tree with $N=2^m-1$ processes (for a large m). All leaves are at the same level.

- How many processes (apart from the sender) receive the query if a child of the root is the querying node, and the $TTL=m-2$?
- What is the minimum TTL required so that all processes receive a query, **no matter who its sender is**?
- Now suppose that, in addition to the tree edges, every process is connected to its immediate left and right neighbors at the same tree level. These links are bidirectional and are present at every level. The links do not wrap around: the leftmost and rightmost processes at a level are not connected to each other. In this topology, what is the minimum TTL required for a query originating at the leftmost leaf to reach all processes?



We recommend that you try to solve the question with small values of m first, and then generalize. We need answers to the above questions in terms of the variable m .

2. Three wealthy industrialists want to make a movie aboard a very unsafe spaceship called "Solargate" that travels to all 8 planets. Lo and behold, the different planets use BitTorrent! While working through an example, you find a case where a file has 7 shards: A, B, C, D, E, F, G. The querying node (peer, leecher) is on a spacecraft, connected to all the planets directly. The planet servers at the 8 planets each have the following shards: Mercury (ACEG), Venus (DF), Earth (ABCDEFG), Mars(BCDEF), Jupiter (ABCDE), Saturn (ABCD), Uranus (ABC), Neptune (AB).

- Which shard will be fetched first?
- Say we introduce a 9th renegade planet. Can you assign a set consisting of just one shard to this 9th planet that changes the first fetched shard by the spacecraft (ties broken by lower letter)?

- c. Suppose the industrialists use a modified version of BitTorrent that always downloads the **most common block first** instead of the rarest block first. What is one problem this could cause?
- d. Suppose they design a modified version of BitTorrent where a file is not split into blocks and a peer downloads the **entire file from a single peer**. What is one problem this could cause compared with BitTorrent's block-based design?

3. You want to make a movie on US Presidential elections. You are putting together the movie's cast. To facilitate the search, you decide to use a Chord-like peer-to-peer system to connect the mobile phones of aspiring artists with each other. The Chord ring has $m = 12$. Nodes with the following peer IDs (or node IDs) join the system: 1992, 1996, 2000, 2004, 2008, 2012, 2016, 2020, 2024, 2028 (those are all years that the Presidential Elections are/will be held). Then, answer the following questions:

- a. Show or list all finger table entries for node 2012.
- b. When all finger tables and successors have converged, show the path taken by a search (or query) message originating from node 2016 intended for the key 1999.
- c. Node 2020 fails. List all nodes whose finger tables need to be updated.

4. A producer, Leo Bloom, likes Bloom filters. He uses his spare time to create a new Bloom filter-like data structure. A (regular) Bloom filter's false positive rate is given as:

$$\left(1 - e^{-\frac{kn}{m}}\right)^k$$

where k is the number of hash functions, n is the size of the input set, and m is the size of the Bloom filter in bits. Leo says that instead of using a single Bloom filter B with 1024 bits and 4 hash functions, his new data structure, called Leo Bloom filter, uses 4 Bloom filters B_1, B_2, B_3 , and B_4 , each with 256 bits, and each using 1 hash function (each hash function is different from the others, and different from the above 4 hash functions). There are two variants: When checking for an item, it returns true only if the item is present in {Variant A: *all*, Variant B: *any*} of B_1, B_2, B_3 , and B_4 . When inserting an item, for both variants A and B, it is inserted into *all* of B_1, B_2, B_3 , and B_4 .

- a. Which of the above three approaches—original using Bloom filter B vs. Leo-any vs. Leo-all—gives the best (lowest) false positive rates? Answer this for two cases: (1) when there are typically 10 elements inserted into the data structure, (2) when there are typically 1000 elements inserted into the data structure. (We recommend that you solve the problem with the variables k, n, m , etc., and then apply these values. But solving with only these two values of 10, 1000, would be ok as well.)

- b. Compare the performance (think about both performance to look up a key and performance to insert a key) of the three approaches.

5. A big-budget movie is being directed by N directors. The N directors never seem to agree on anything. To solve this, the producer decides to implement a quorum approach. In a system of N processes (N large enough), THREE quorum sets Q_1 , Q_2 , and Q_3 are selected in an arbitrary manner, all of the same size M .

- a. What should this minimum value of M be so that at least r processes belong to ALL three of Q_1 , Q_2 , and Q_3 (that is, the intersection of all three sets is r)? r is a constant, much smaller than N .
- b. Solve the problem again instead now with k quorum sets $Q_1, Q_2, Q_3, \dots, Q_k$, where k is a constant that is much smaller than N .

(Hint: start with 2 sets intersecting at one process, i.e., $k=2$ and $r=1$, and quantitatively reason/think about why it has to be $N/(2+1)$).

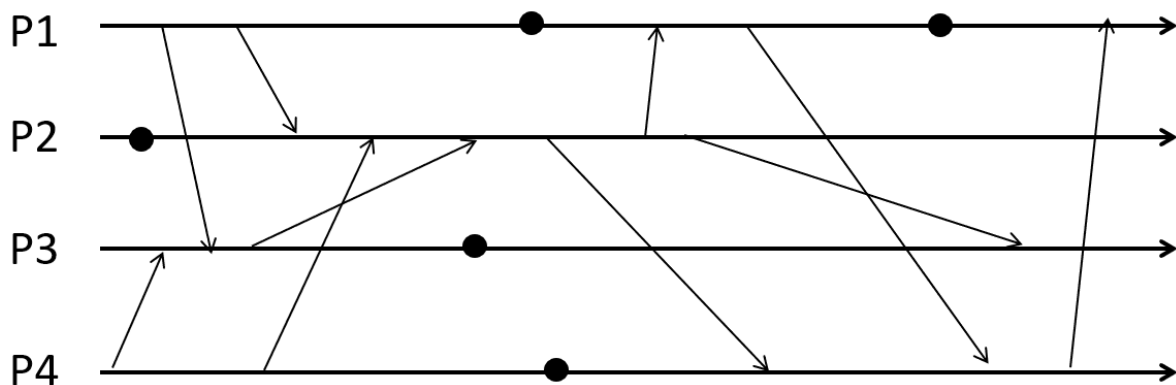
6. The director C. Nolan likes to deal with time travel, which means he proposes a lot of “What if?” proposals. He asks you several questions about Cassandra’s design. For each of the “what if” questions, give at least one reason arguing against his proposal.

- a. What if Cassandra uses finger tables from Chord instead of full membership lists?
- b. What if Cassandra didn’t use tombstones for deletes and instead deleted records directly from Memtables and SSTables (but the rest of the system works the same way)?
- c. What if there were no Bloom filters (but the rest of the system works the same way)?
- d. What if, for an update, a replica first checked whether the key exists in an SSTable on disk and modified the SSTable directly if it did; only if the key did not exist in any SSTable would the replica insert the update into the Memtable? (but the rest of the system works the same way)?

7. Katherine Johnson was a Black computer mathematician and scientist who worked at NASA. Being one of the first to write computer programs involving space and time, the director of a movie about her is looking into Cristian’s algorithm. They find that the round-trip time for one round of synchronization messages between a client and a server is exactly 1.01 ms. They are trying to calculate the error for this run, so they calculate some minimum delays. On the server side, they find that there is a delay of at least 15.7 microseconds for a packet to get from an application to the network interface,

and the delay in the opposite path (network interface to application buffer) is at least 0.4 ms. On the client side, they measure that the minimum time to get from the network interface to the application buffer to be at least 100 microseconds, and the minimum time on the reverse path as X microseconds. But they forget to measure X. What is the error, given the data just presented?

8. Spiderman and his doppelgangers in the metaverse are trying to figure out the communication among the different metaverses. They have the following timeline of messages exchanged, where each dot represents an instruction. Can you mark *Lamport timestamps* on each event? It is ok to print this out and hand-draw/write the timestamps then scan or photograph your solution, and insert it into your solution doc.



9. Whoops, the simultaneous and sudden arrival of all the concurrent villains (Doc Ock, Green Goblin, Electro, Sandman, and the others) means Lamport timestamps are unusable for distinguishing concurrent events from causal events. Can you mark *vector timestamps* for the timeline from the previous question? It is ok to print this out and hand-draw/write the timestamps then scan or photograph your solution, and insert it into your solution doc.

10. The movie is a hit! The breakout star of the movie, Kristen, was so happy with your work that she has asked the production company to give you one last puzzle to solve before you can be paid the millions of \$ you are owed.

The puzzle concerns timestamps.

- a. Consider a modified version of the *Lamport timestamp* algorithm. Instead of initializing every process's local counter (clocks) to 0, each process initializes its counter to a *random* positive integer (randomly selected for each process). The rest of the algorithm remains unchanged. Would these new timestamps still

preserve (obey) causality, i.e., guarantee that $E1 \rightarrow E2 \Rightarrow \text{timestamp}(E1) < \text{timestamp}(E2)$? Explain why or why not.

b. Consider a modified version of the *Vector timestamp* algorithm. Instead of initializing every process's local clocks to a vector of all 0s (i.e., $[0,0,..0]$), each process initializes its clocks to a vector of *random* positive integers (each integer for each process independently selected). The rest of the algorithm remains unchanged. Would these new timestamps still preserve (obey) causality, i.e., guarantee that $E1 \rightarrow E2 \Rightarrow \text{timestamp}(E1) < \text{timestamp}(E2)$? Explain why or why not.

c. Would the algorithm in (b) still preserve all the properties of vector timestamps? Explain why or why not.